

Education			
Year	Degree/Certificate	Institute	CPI/%
Oct 2022 – Aug 2026	B.Tech in CS & IT	Ajay Kumar Garg Engineering College, Delhi NCR	7.25/10
2021 – 2022	Senior Secondary (XII)	New Era School, Delhi NCR	91.2%
2017 – 2020	High School (X)	SJN International School, Coimbatore	92.6%

Experience

Google Summer of Code 2025 Contributor | *P4 Language Consortium, Stanford, CA* (May 2025 – Sept 2025)

SpliDT Project: Scaling Stateful Decision Tree Algorithms in P4

P4, Python, P4Runtime, Docker, C++, Intel Tofino

- Switch-Native Architecture:** Architected a framework to compile ML decision trees directly into P4 data planes, implementing Partitioned Decision Trees (PDTs) with TCAM-efficient Subtree IDs (SIDs) to resolve hardware ASIC constraints.
- Validation & Optimization:** Engineered a Jinja2-based P4Runtime testing pipeline (Tofino/Mininet) improving deployment reliability by 30%, and optimized SID recirculation to achieve 10 Gbps line-rate throughput with 20% lower inference latency.

Open Source Contributions

- WasmEdge/WasmEdge:** #4470 fix(wasi): resolve 100% CPU usage on macOS due to clock mismatch; #4466 feat(docker): add standalone alpine-base image build pipeline.
- Project-HAMi/HAMi:** #2292 fix(device): use len(devices) for GenReason in kunlun and awsneuron; #2307 fix(device-plugin): release node lock when bind-phase patch fails; #2259 fix(scheduler): handle ResourceQuota tombstones; #2204 fix(metrics): prevent invalid memory ratios for unknown total memory; #2221 fix(workflow): skip non-translatable issue comments.
- facebook/bpfilter:** #507 build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c.

Key Projects

Cutable | *Go, Next.js, PostgreSQL, OpenRouter, E2B, WebSockets, Nomad* (Project Link)

Objective:

Approach:

Impact:

DMIE | *Next.js, TypeScript, Python, FastAPI, Vector DB, LLM* (Project Link)

Objective:

Approach:

Impact:

Slix | *React, Go, Azure OpenAI, REST API* (Project Link)

Objective:

Approach:

Impact:

Myprod | *Go, Nomad, Traefik, WireGuard, Docker, Netlify DNS, SOPS* (Project Link)

Objective:

Approach:

Impact:

XNIC v1 | *Linux Kernel, C, PCI, BAR0 MMIO, DMA, Descriptor Rings, e1000, Intel 82540EM, net_device, NAPI, IRQ, MSI, ethtool, QEMU, HVF, KFENCE, sparse, DPDK, rte_ethdev, net_pcap PMD, tcpdump, Wireshark, W5500, SPI, Device Tree, Bash, GitHub Actions, Vercel* (Project Link)

Objective:

Approach:

Impact:

Technical Skills

- Language:** Python, C, C++, JavaScript, TypeScript, Go, P4
- Libraries/Frameworks:** NodeJS, Numpy, Matplotlib, eBPF, gRPC, Protobufs, Next.js, Scapy, MLIR, LLVM, Sklearn, React, Web Services
- Tools/Platforms:** Docker, CI/CD, VS Code, UTM, GitHub, GitLab, DaVinci Resolve, Linux (x86_64, ARM64 & AMD64)
- Databases:** SQL, MongoDB, GraphQL, Redis, PostgreSQL, MySQL

Positions of Leadership

Technical Writer, Cloud Native Community Group, New Delhi (May 2024 – Dec 2024)

- Led a team of 6+ content writers and 4 designers to deliver documentation and event management support.
- Partnered with marketing to revamp social media presence across LinkedIn, Instagram, and X, achieving a 60% bump in engagement.

- **Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (Nov 2023 – Nov 2025)
 - Co-led **10+** tech events & anchored the GDSC WOW flagship event at IIIT Delhi, with the **31+** crew members.
 - Engaged a live audience of **800+** at WOW and **500+** developers overall, boosting participation by **30%** and visibility by **40%**.

Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, “Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions,” *2026 International Conference on Computing, Sciences and Communications (ICCSC)*, Ghaziabad, India, Feb. **2026**.
[IEEE Xplore](#)  [DOI: 10.1109/ICCSC67078.2026.11468597](#)

Certifications & Honors

Certifications	• Open Science 101 – NASA • Data Science for Engineers – NPTEL • Enhancing Soft Skills & Personality – NPTEL • IIRS Satellite Remote Sensing – ISRO
Honors & Awards	• 2x AKTU State Men Singles TT Gold (2023, 2024) • International Robotics Olympiad Bronze, Thailand (2019) • Merit Certification at Carnic Model United Nation (2021) • NCC Cadet, National Cadet Corps (2019)